DSA INTERVIEW MASTER PROBLEM BANK

JAVA EDITION • BEGINNER TO ADVANCED • COMPANY-AGNOSTIC CORE

Purpose

A structured practice bank mapped to the topics and algorithms in your uploaded 9-page DSA syllabus. It combines core interview patterns, representative named problems, and selected advanced algorithm exercises.

How to use this bank

1. Solve HIGH priority problems first, without looking at solutions.
2. For every problem, write the brute-force idea, optimal algorithm, proof/intuitive reasoning, time complexity and space complexity.
3. Re-solve missed problems after 2 days, 1 week and 3 weeks.
4. Implement the underlying data structure/algorithm in Java, not just the individual answer.
5. Attempt MEDIUM and then selected LOW problems after core patterns are strong.

Important accuracy note

No public source contains every interview question ever asked, and future questions cannot be predicted with certainty. Company-tagged question databases are candidate-reported and vary by time, role and location. This is a curated preparation bank—not a guarantee, leaked question list, or claim that every listed problem was asked by a named company.

Difficulty labels

Easy / Medium / Hard are approximate platform-style labels. “Core” and “Advanced” identify classic algorithm exercises that may not have a single LeetCode ID.

Coverage overview

Total curated entries: **218** HIGH: **184** MEDIUM: **16** LOW: **18**

1. Arrays, Hashing & Prefix Sum — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Two Sum	1	Easy	HashMap / complement lookup
Contains Duplicate	217	Easy	HashSet
Valid Anagram	242	Easy	Frequency counting
Group Anagrams	49	Medium	Hashing / canonical key
Top K Frequent Elements	347	Medium	HashMap + heap / bucket sort
Product of Array Except Self	238	Medium	Prefix and suffix products
Longest Consecutive Sequence	128	Medium	HashSet; sequence start detection
Subarray Sum Equals K	560	Medium	Prefix sum + HashMap
Maximum Subarray	53	Medium	Kadane's algorithm
Maximum Product Subarray	152	Medium	DP with min/max ending product
Best Time to Buy and Sell Stock	121	Easy	One-pass minimum tracking
Best Time to Buy and Sell Stock II	122	Medium	Greedy local gains
Majority Element	169	Easy	Boyer-Moore voting
Sort Colors	75	Medium	Dutch National Flag / 3 pointers
Merge Sorted Array	88	Easy	Two pointers from end
Rotate Array	189	Medium	Reverse sections / cyclic rotation
Find All Numbers Disappeared in an Array	448	Easy	Index marking
First Missing Positive	41	Hard	Cyclic placement / index mapping
Set Matrix Zeroes	73	Medium	In-place marker rows/columns
Spiral Matrix	54	Medium	Boundary simulation
Subarray Product Less Than K	713	Medium	Variable sliding window
Continuous Subarray Sum	523	Medium	Prefix remainder + HashMap

2. Two Pointers & Sliding Window — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Valid Palindrome	125	Easy	Two pointers
Two Sum II - Input Array Is Sorted	167	Medium	Opposite-direction pointers
3Sum	15	Medium	Sorting + two pointers
3Sum Closest	16	Medium	Sorting + two pointers
4Sum	18	Medium	Sorting + nested two pointers
Container With Most Water	11	Medium	Two pointers / greedy elimination
Trapping Rain Water	42	Hard	Two pointers or monotonic stack
Longest Substring Without Repeating Characters	3	Medium	Variable sliding window + set/map
Longest Repeating Character Replacement	424	Medium	Sliding window + frequency
Permutation in String	567	Medium	Fixed window + frequency
Minimum Window Substring	76	Hard	Variable window + counts
Sliding Window Maximum	239	Hard	Monotonic deque
Minimum Size Subarray Sum	209	Medium	Variable sliding window
Max Consecutive Ones III	1004	Medium	Window with at most K zeros

3. Binary Search & Sorting — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Binary Search	704	Easy	Classic binary search
Search Insert Position	35	Easy	Lower bound
Find First and Last Position of Element in Sorted Array	34	Medium	Two boundary searches
Search in Rotated Sorted Array	33	Medium	Modified binary search
Find Minimum in Rotated Sorted Array	153	Medium	Modified binary search
Search a 2D Matrix	74	Medium	Binary search on flattened matrix
Koko Eating Bananas	875	Medium	Binary search on answer
Capacity To Ship Packages Within D Days	1011	Medium	Binary search on answer
Median of Two Sorted Arrays	4	Hard	Binary search on partition
Find Peak Element	162	Medium	Binary search on slope
Sort an Array	912	Medium	Merge sort / heap sort / quicksort
Kth Largest Element in an Array	215	Medium	Quickselect / heap
Count of Smaller Numbers After Self	315	Hard	Merge sort counting / Fenwick tree

4. Linked Lists — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Reverse Linked List	206	Easy	Iterative / recursive pointer reversal
Merge Two Sorted Lists	21	Easy	Dummy node + two pointers
Linked List Cycle	141	Easy	Floyd slow/fast pointers
Linked List Cycle II	142	Medium	Floyd cycle-entry proof
Middle of the Linked List	876	Easy	Slow/fast pointers
Remove Nth Node From End of List	19	Medium	Two pointers + dummy node
Reorder List	143	Medium	Find middle + reverse + merge
Add Two Numbers	2	Medium	Digit simulation + carry
Intersection of Two Linked Lists	160	Easy	Pointer switching
Palindrome Linked List	234	Easy	Middle + reverse half
Copy List with Random Pointer	138	Medium	HashMap / interweaving
Reverse Nodes in k-Group	25	Hard	Segment reversal
LRU Cache	146	Medium	HashMap + doubly linked list

5. Stack, Queue & Monotonic Structures — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Valid Parentheses	20	Easy	Stack
Min Stack	155	Medium	Auxiliary stack / encoded minimum
Evaluate Reverse Polish Notation	150	Medium	Stack evaluation
Daily Temperatures	739	Medium	Monotonic decreasing stack
Next Greater Element I	496	Easy	Monotonic stack + map
Largest Rectangle in Histogram	84	Hard	Monotonic increasing stack
Asteroid Collision	735	Medium	Stack simulation
Decode String	394	Medium	Stack / nested parsing
Implement Queue using Stacks	232	Easy	Amortized two-stack queue
Sliding Window Maximum	239	Hard	Monotonic deque
Basic Calculator	224	Hard	Stack + expression parsing
Trapping Rain Water	42	Hard	Monotonic stack alternative

6. Trees & Binary Search Trees — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Maximum Depth of Binary Tree	104	Easy	DFS recursion / BFS
Same Tree	100	Easy	Recursive tree comparison
Invert Binary Tree	226	Easy	DFS / BFS
Binary Tree Level Order Traversal	102	Medium	BFS queue
Binary Tree Zigzag Level Order Traversal	103	Medium	BFS + level direction
Diameter of Binary Tree	543	Easy	Postorder DFS
Balanced Binary Tree	110	Easy	Postorder height check
Lowest Common Ancestor of a Binary Tree	236	Medium	Postorder recursion
Validate Binary Search Tree	98	Medium	Range constraints / inorder
Kth Smallest Element in a BST	230	Medium	Inorder traversal
Construct Binary Tree from Preorder and Inorder Traversal	105	Medium	Divide and conquer + index map
Binary Tree Right Side View	199	Medium	BFS / DFS by depth
Path Sum	112	Easy	DFS path tracking
Binary Tree Maximum Path Sum	124	Hard	Postorder DP
Serialize and Deserialize Binary Tree	297	Hard	DFS/BFS encoding
Flatten Binary Tree to Linked List	114	Medium	Reverse preorder / pointer rewiring
Subtree of Another Tree	572	Easy	Tree traversal + equality
Recover Binary Search Tree	99	Medium	Inorder inversion detection

7. Heaps & Priority Queues — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Kth Largest Element in an Array	215	Medium	Min heap / quickselect
Top K Frequent Elements	347	Medium	Heap / bucket sort
K Closest Points to Origin	973	Medium	Max heap / quickselect
Find Median from Data Stream	295	Hard	Two heaps
Merge k Sorted Lists	23	Hard	Min heap / divide and conquer
Task Scheduler	621	Medium	Greedy + max heap / counting
Meeting Rooms II	253	Medium	Min heap of end times
Find K Pairs with Smallest Sums	373	Medium	Heap over pair frontier

8. Graphs: Traversal, Topological Sort & DSU — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Number of Islands	200	Medium	Grid DFS/BFS
Max Area of Island	695	Medium	Grid DFS/BFS
Clone Graph	133	Medium	BFS/DFS + visited map
Rotting Oranges	994	Medium	Multi-source BFS
Course Schedule	207	Medium	Cycle detection / topological sort
Course Schedule II	210	Medium	Kahn's algorithm / DFS topo
Graph Valid Tree	261	Medium	DFS/BFS or DSU
Number of Connected Components in an Undirected Graph	323	Medium	DFS/BFS / DSU
Redundant Connection	684	Medium	Disjoint Set Union
Is Graph Bipartite?	785	Medium	BFS/DFS two-coloring
Pacific Atlantic Water Flow	417	Medium	Reverse multi-source DFS/BFS

Problem	LC ID	Level	Pattern / algorithm to practise
Word Ladder	127	Hard	BFS shortest path
Network Delay Time	743	Medium	Dijkstra + priority queue
Cheapest Flights Within K Stops	787	Medium	Bellman-Ford / bounded relaxation
Path With Minimum Effort	1631	Medium	Dijkstra / binary search + BFS
Min Cost to Connect All Points	1584	Medium	Prim / Kruskal MST
Accounts Merge	721	Medium	DSU
Evaluate Division	399	Medium	Weighted graph DFS/BFS

9. Greedy & Intervals — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Merge Intervals	56	Medium	Sort + interval merge
Insert Interval	57	Medium	Interval scan
Non-overlapping Intervals	435	Medium	Greedy by earliest finish
Meeting Rooms	252	Easy	Sort intervals
Meeting Rooms II	253	Medium	Sweep line / min heap
Jump Game	55	Medium	Greedy reachability
Jump Game II	45	Medium	Greedy range expansion
Gas Station	134	Medium	Greedy total and reset
Partition Labels	763	Medium	Last occurrence + greedy
Task Scheduler	621	Medium	Greedy + frequency
Hand of Straights	846	Medium	Ordered counts / greedy
Candy	135	Hard	Two-pass greedy
Huffman Coding (construct and decode)	—	Advanced	Greedy + priority queue; implement as a design exercise

10. Backtracking & Recursion — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Subsets	78	Medium	Choose-explore-unchoose
Subsets II	90	Medium	Backtracking + duplicate pruning
Permutations	46	Medium	Backtracking + used array
Permutations II	47	Medium	Sorting + duplicate pruning
Combination Sum	39	Medium	Backtracking with reuse
Combination Sum II	40	Medium	Backtracking without reuse
Generate Parentheses	22	Medium	Constraint-guided backtracking
Word Search	79	Medium	Grid DFS + backtracking
Palindrome Partitioning	131	Medium	Backtracking + palindrome check
N-Queens	51	Hard	Backtracking + column/diagonal sets
Sudoku Solver	37	Hard	Constraint propagation + backtracking
Letter Combinations of a Phone Number	17	Medium	Cartesian product / backtracking
Restore IP Addresses	93	Medium	Bounded choice backtracking

11. Dynamic Programming — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Climbing Stairs	70	Easy	1D DP / Fibonacci
House Robber	198	Medium	1D DP / take-or-skip
House Robber II	213	Medium	Circular DP split

Problem	LC ID	Level	Pattern / algorithm to practise
Coin Change	322	Medium	Unbounded knapsack / min DP
Coin Change II	518	Medium	Unbounded knapsack / count ways
Partition Equal Subset Sum	416	Medium	0/1 knapsack boolean DP
Target Sum	494	Medium	Subset-sum transformation
Longest Increasing Subsequence	300	Medium	DP / patience sorting
Longest Common Subsequence	1143	Medium	2D sequence DP
Edit Distance	72	Medium	2D string DP
Word Break	139	Medium	Prefix DP + set
Unique Paths	62	Medium	Grid DP
Minimum Path Sum	64	Medium	Grid DP
Decode Ways	91	Medium	1D DP
Maximum Product Subarray	152	Medium	DP tracking min/max
Palindromic Substrings	647	Medium	Expand around center / DP
Longest Palindromic Subsequence	516	Medium	Interval DP
Burst Balloons	312	Hard	Interval DP
Matrix Chain Multiplication	—	Advanced	Interval DP; implement classic recurrence
Best Time to Buy and Sell Stock with Cooldown	309	Medium	State-machine DP
Regular Expression Matching	10	Hard	2D string DP
Distinct Subsequences	115	Hard	2D counting DP
Triangle	120	Medium	Bottom-up DP / space optimization
Maximal Square	221	Medium	2D DP
Binary Tree Maximum Path Sum	124	Hard	Tree DP

12. Bit Manipulation & Math — HIGH

Problem	LC ID	Level	Pattern / algorithm to practise
Single Number	136	Easy	XOR cancellation
Number of 1 Bits	191	Easy	Brian Kernighan / bit count
Counting Bits	338	Easy	DP on bits
Reverse Bits	190	Easy	Bit extraction and placement
Missing Number	268	Easy	XOR / arithmetic sum
Sum of Two Integers	371	Medium	XOR sum + carry
Power of Two	231	Easy	n & (n-1)
Bitwise AND of Numbers Range	201	Medium	Common-prefix bit shifting
Single Number II	137	Medium	Bit counts modulo 3
Subsets	78	Medium	Bitmask enumeration
Pow(x, n)	50	Medium	Binary exponentiation
Greatest Common Divisor (Euclidean algorithm)	—	Core	Euclidean algorithm; implement
Count Primes	204	Medium	Sieve of Eratosthenes
Happy Number	202	Easy	Cycle detection / digit transform
Factorial Trailing Zeroes	172	Medium	Prime factor counting

13. Tries, Advanced Graphs & Range Queries — MEDIUM

Problem	LC ID	Level	Pattern / algorithm to practise
Implement Trie (Prefix Tree)	208	Medium	Trie insert/search/startsWith
Design Add and Search Words Data Structure	211	Medium	Trie + DFS wildcard
Word Search II	212	Hard	Trie + grid backtracking

Problem	LC ID	Level	Pattern / algorithm to practise
Implement Prefix Sum / Range Sum Query - Immutable	303	Easy	Prefix sums
Range Sum Query - Mutable	307	Medium	Segment tree / Fenwick tree
Count of Smaller Numbers After Self	315	Hard	Fenwick tree / merge sort
Range Sum Query 2D - Immutable	304	Medium	2D prefix sum
Shortest Path in Binary Matrix	1091	Medium	BFS
Critical Connections in a Network	1192	Hard	Tarjan bridges / low-link
Strongly Connected Components	—	Advanced	Kosaraju / Tarjan; implement on directed graph
Eulerian Path and Circuit	—	Advanced	Hierholzer's algorithm; implement
Shortest Path in a DAG	—	Advanced	Topological order + relaxation
0-1 BFS	—	Advanced	Deque-based shortest path
Floyd-Warshall	—	Advanced	All-pairs shortest paths
Bellman-Ford	—	Advanced	Repeated edge relaxation
Articulation Points	—	Advanced	DFS discovery/low-link

14. Advanced / Low-frequency extensions — LOW

Problem	LC ID	Level	Pattern / algorithm to practise
Palindrome Pairs	336	Hard	Trie / hashing + string reversal
Longest Duplicate Substring	1044	Hard	Binary search + rolling hash
Find the Shortest Superstring	943	Hard	Bitmask DP
Minimum Cost to Merge Stones	1000	Hard	Interval DP
Remove Boxes	546	Hard	3D interval DP
Shortest Path Visiting All Nodes	847	Hard	Bitmask BFS
Smallest Sufficient Team	1125	Hard	Bitmask DP
Swim in Rising Water	778	Hard	Dijkstra / binary search + BFS
Reconstruct Itinerary	332	Hard	Eulerian path + Hierholzer
Max Flow / Min Cut (classic network-flow exercise)	—	Advanced	Ford-Fulkerson / Edmonds-Karp
Bipartite Matching (classic assignment exercise)	—	Advanced	Augmenting paths / Hopcroft-Karp
Manacher's Algorithm (longest palindromic substring)	—	Advanced	Linear-time palindrome radii
Suffix Array + LCP (classic string exercise)	—	Advanced	Suffix sorting + LCP construction
Digit DP (count numbers in range with digit constraints)	—	Advanced	Digit-position / tight-state DP
Segment Tree with Lazy Propagation (range add, range sum)	—	Advanced	Lazy segment tree
Sparse Table (static range minimum query)	—	Advanced	Idempotent range query
Reservoir Sampling (streaming k-sample)	—	Advanced	Uniform sampling from stream
Closest Pair of Points	—	Advanced	Divide and conquer geometry

Algorithm implementation checklist

Arrays / strings: Two pointers, sliding window, prefix/suffix sums, difference array, Kadane, sorting, matrix traversal

Searching / sorting: Binary search boundaries, binary search on answer, merge sort, quicksort, heap sort, quickselect

Linked list: Pointer reversal, Floyd cycle detection, merge, dummy node, fast/slow pointer, LRU design

Stack / queue: Monotonic stack, monotonic deque, expression evaluation, circular queue

Trees / heaps / tries: DFS traversals, BFS, BST invariants, heap operations, top-K, trie operations

Graphs: BFS, DFS, multi-source BFS, topological sort, DSU, Dijkstra, Bellman-Ford, Floyd-Warshall, Prim, Kruskal, SCC, bridges

Recursion / backtracking: Choose-explore-unchoose, pruning, duplicate handling, constraint ordering

Greedy: Interval scheduling, exchange reasoning, sweep line, Huffman coding

Dynamic programming: State, transition, base case, iteration order, memoization, tabulation, space optimization, knapsack, interval, tree and bitmask DP

Bit / math: XOR, bit masks, Brian Kernighan, Euclidean GCD, sieve, modular arithmetic, binary exponentiation

Advanced structures: Fenwick tree, segment tree, lazy propagation, sparse table, suffix array, rolling hash

Sources and verification approach

The uploaded syllabus is the structural basis for topic coverage. The problem selection is cross-checked against public curated interview-preparation plans and candidate-reported company question indexes. These sources are useful for breadth and trend signals, but they do not establish a complete historical record or guarantee future questions.

LeetCode Top Interview 150: https://leetcode.com/studyplan/top-interview-150/
NeetCode 150 roadmap: https://neetcode.io/roadmap
Company-wise candidate-reported questions: https://crackedprep.com/company-wise-leetcode-questions

Practice note: A problem is complete only when you can explain the pattern, derive the solution, code it in Java, test edge cases, and state complexity without assistance.